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Hannah Scholz

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Advisor: Dr. Michael Rothgang

Second Advisor: Prof. Dr. Floris van Doorn

MATHEMATISCHES INSTITUT

MATHEMATISCH-NATURWISSENSCHAFTLICHE FAKULTÄT DER
RHEINISCHEN FRIEDRICH-WILHELMS-UNIVERSITÄT BONN

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1 The mathematics of collar neighborhoods

1.1 Basics about collar neighbourhoods

Definition 1.1.1. Let M be a $\mathcal{C}^k$ -manifold with boundary and let U be a neighbourhood of the boundary ∂M . U is called a *collar neighbourhood* if there is a $\mathcal{C}^k$ -diffeomorphism $f: \partial M \times [0, 1) \rightarrow U$ s.t. for all $x \in \partial M$, $f(x, 0) = x$.

